

First Terminal Examination - 2080

Class : XI

Subject: Biology (Code: 2011)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet. [6 × 1 = 6]

1. The middle lamella of cell wall is composed of
 - a. Calcium & Magnesium pectate
 - b. Cellulose, hemicellulose & pectin
 - c. Peptidoglycan
 - d. Lignin
2. Which one of the following forms primary lysosome?
 - a. Ribosome
 - b. Golgi body
 - c. Endoplasmic reticulum
 - d. Mitochondria
3. The two unequal subunits of ribosome remain attached together by means of
 - a. Hydrogen bond
 - b. mRNA
 - c. Binding force of magnesium atom
 - d. Polyribosome
4. What is the distinctive feature of *Spirogyra* that gives it's name?
 - a. Presence of Mucilage
 - b. Fresh water habitat
 - c. Multiple flagella
 - d. Spiral shaped chloroplast
5. In binomial nomenclature, every organism has
 - a. Two names, one Latin, other common
 - b. Two names, genus & species
 - c. Two names, one scientific, other common
 - d. Two names by two scientists.
6. The taxonomic group present below the class is
 - a. Division
 - b. Family
 - c. Order
 - d. Phylum

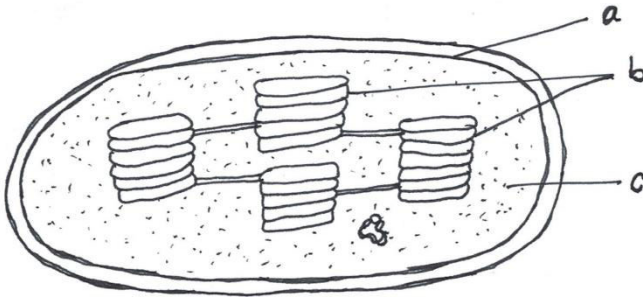
Group 'B' [BOTANY]

Give short answer to the following questions.

[4 × 4 = 16]

1. Study the following diagram & answer the questions.

[1×4 = 4]



- a) Name the above figure.
b) Name the parts a, b & c.
2. Who proposed cell theory? Write down the postulations of cell theory. [1+3]

OR

- What is the nature of plasmamembrane? Write down its functions. [1+3]
3. Why bryophytes are called amphibians of plant kingdom? Write down general characters of bryophytes. [1+3]
4. What is the function of Pyrenoid? Draw a well labelled diagram to show the cell structure of *Spirogyra* (no description). [1+3]

Group 'C' [BOTANY]

Give long answer to the following questions.

[8 × 2 = 16]

5. What are cell organelles? Why mitochondria are called power house of the cell? Describe the structure & functions of mitochondria. [1+1+3+3]
6. What is the function of gemma cup? *Marchantia* is dioecious plant what does it mean? Describe sexual reproduction in *Marchantia*. [1+1+6]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet. [5 × 1 = 5]

1. Ophthalmology is the study of
 - a) Eye
 - b) Teeth
 - c) Tongue
 - d) Brain
2. Branch of Zoology which deals with mechanisms of heredity is called
 - a) Genetics
 - b) Eugenics
 - c) Euthenics
 - d) Cytology
3. The mutation theory of evolution was proposed by :
 - a) Darwin
 - b) Lamarck
 - c) Hugo de Vries
 - d) Huchingson
4. Charles Darwin's theory of evolution is based on:
 - a) Environment
 - b) Chromosome pairing
 - c) Mutation
 - d) Natural Selection
5. The organs which have same basic structure and origin but differ in function are called as:
 - a) Homologous organs
 - b) Analogous organs
 - c) Vestigial organs
 - d) None

Group 'B' [ZOOLOGY]

Give short answers to the following questions.

[4 × 4 = 16]

1. How biology is related with chemistry and physics? [2+2]
2. Summarize Miller's simulation experiment for organic synthesis with figure. Comment on its efficacy. [2+2]
3. "Biology is the science of exception". Elaborate this statement. [4]
4. *Archaeopteryx lithographica* is missing link between reptiles and aves, discuss. [2+2]

Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

5. How Neo-Darwinism explains the evolution of life? Explain. [2+6]
6. Explain the theory of "Inheritance of acquired characters" in your own words. [2+6]

ॐ The End ॐ